



Centre for
Robotics



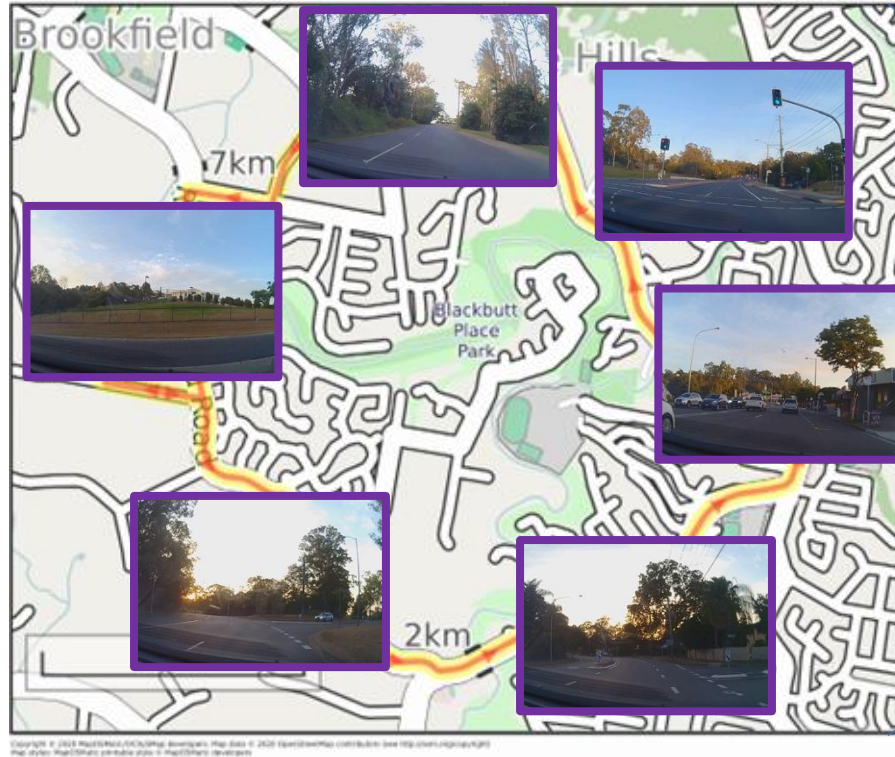
Matched Filter based LiDAR Place Recognition for Urban and Natural Environments

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Place recognition

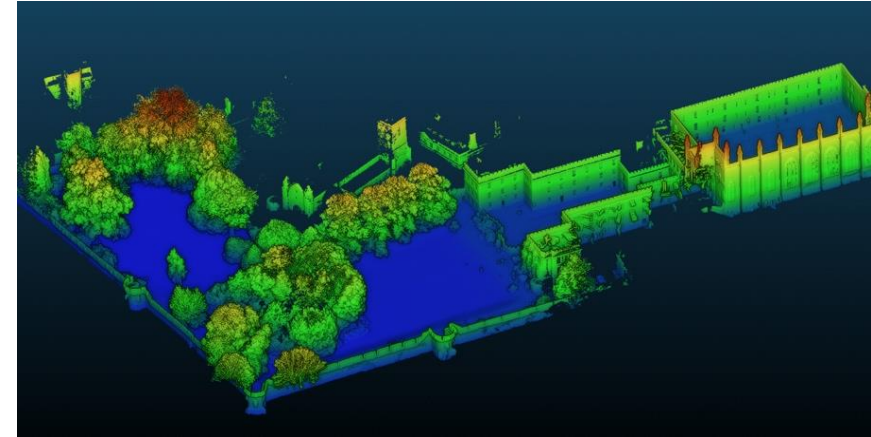
Reference Database



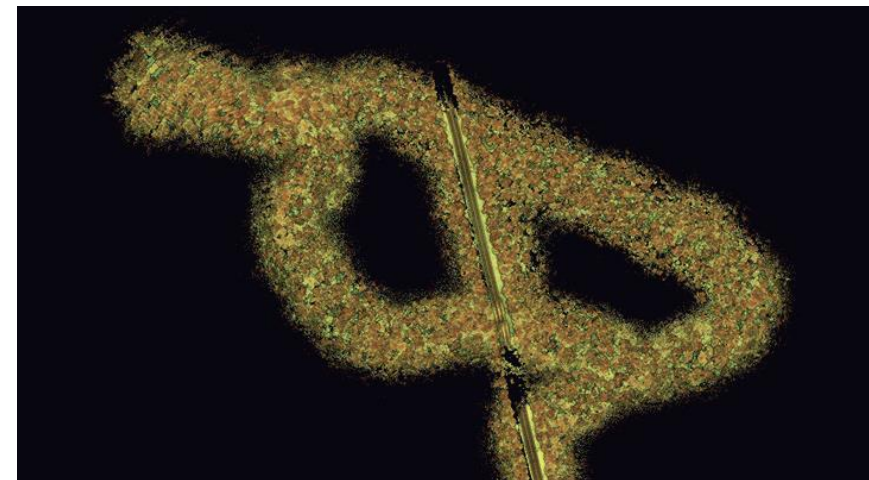
- Place recognition is required for loop closure and also enables global re-localisation in GPS-denied conditions.
- It compares the current query location against a reference database to find a match.

Existing Methods for LiDAR place recognition

- Handcrafted descriptors [1,2,3,4] compress 3D scans into compact, rotation-tolerant representations.
- Learned descriptors [5,6,7,8] achieves higher recall and robustness by training on large-scale LiDAR datasets from structured urban settings with stable features like buildings, poles, and road edges.
- Performance drops in unstructured or natural environments (forests, trails, foliage) where geometry is irregular and changes over time.



LiDAR map from a structured environment (Oxford New College)



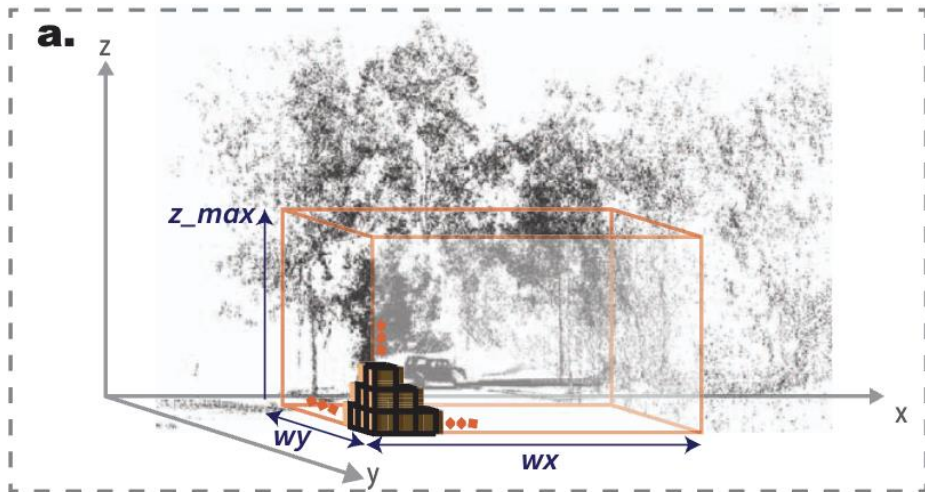
LiDAR map from an unstructured environment (WildPlaces)

[1,2,3,4] He et al., IROS 2016 (M2DP); Kim & Kim, IROS 2018 (Scan Context); Xu et al., TRO 2022 (RING++); Wang et al., IROS-2020 (LiDAR-Iris)

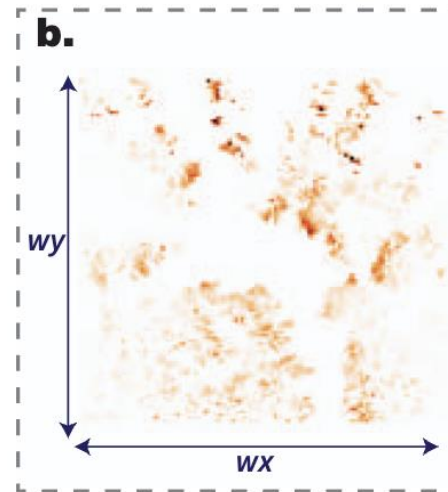
[5,6,7,8] Uy et al., CVPR 2018 (PointNetVLAD), Zhang et al., CVPR 2019 (PCAN), Liu et al., ICCV 2019 (LPD-Net), Du et al., CVPR 2020 (DH3D)

Method: Generating 2D descriptors

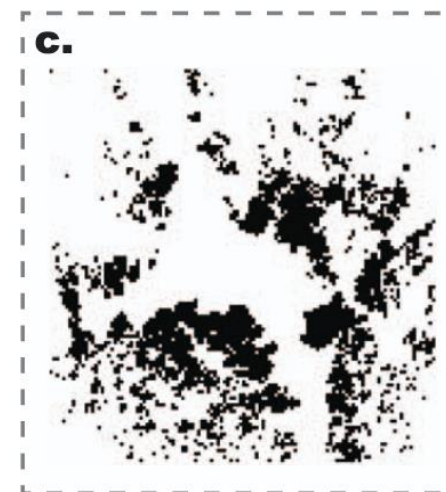
To handle similarity search across large point clouds, the 3D point cloud is transformed into a 2D birds eye view descriptor



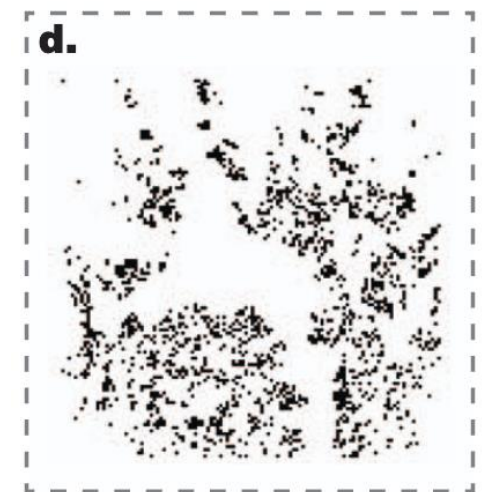
Voxelised Point Cloud



High Density Map

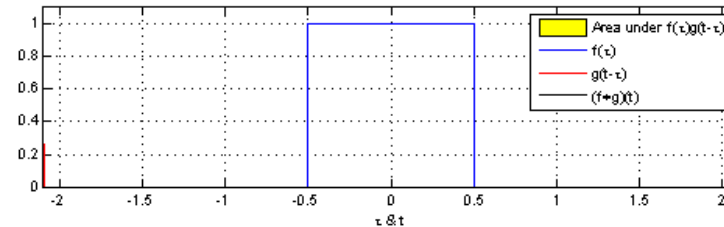


Occupancy Grid



Downsampled Occupancy Grid

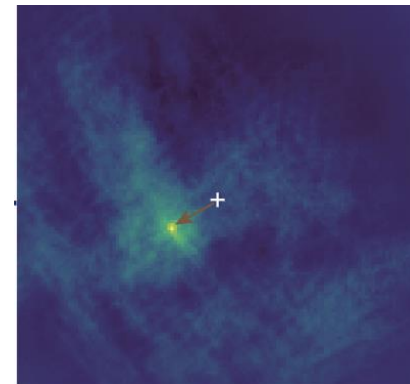
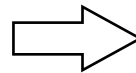
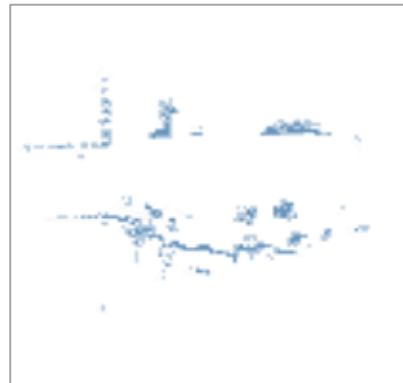
Method: Matched Filtering with LiDAR BEV Descriptors



Matched filter from signal processing : Detecting a known signal amidst noise

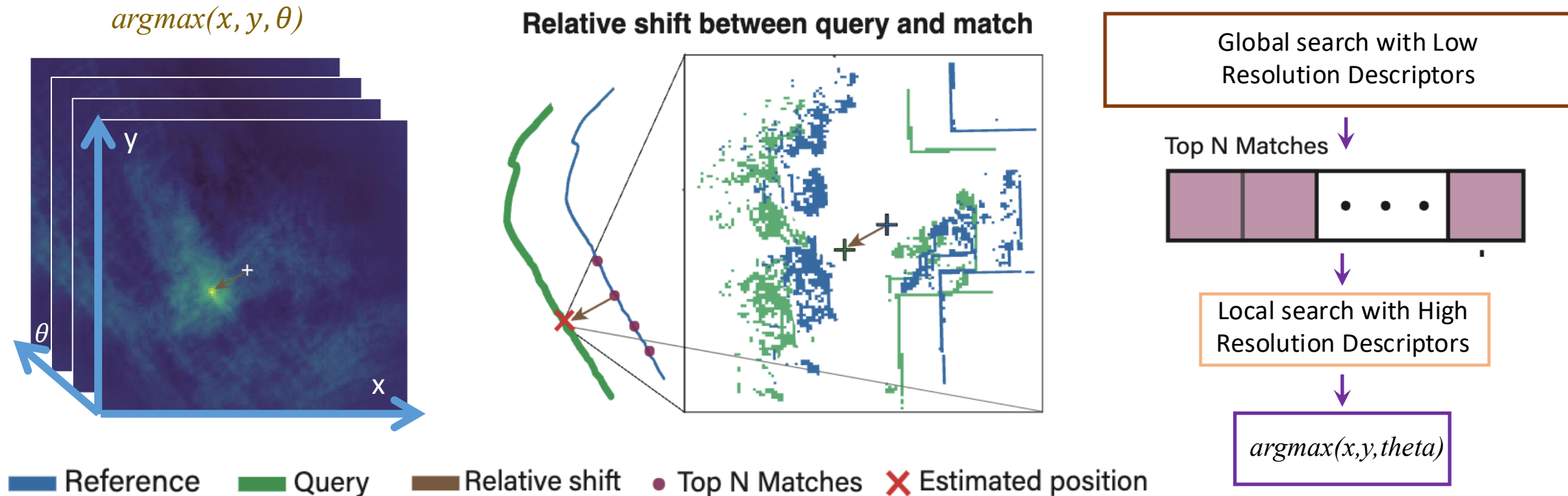


$$(Q * R)(i_r, j_r, \theta) = \sum_{i_q} \sum_{j_q} Q(i_q, j_q, \theta) \cdot R(i_q + i_r, j_q + j_r, 0)$$

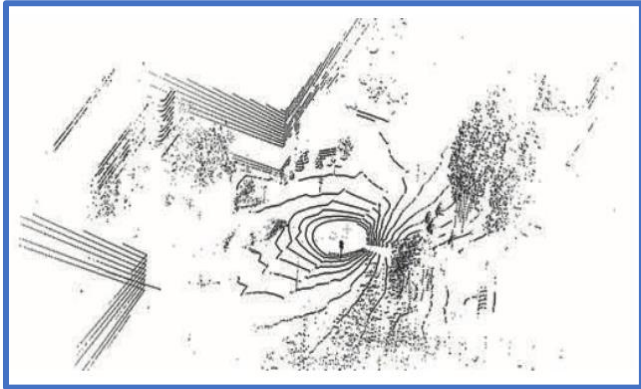


Finding a matched place with rotated 2D lidar descriptors convolved against the reference database

Method: Relative Pose Estimation and Two-Stage Search



Datasets Evaluated



NCLT [1]

- 5.5km Route
- 2 Reference traverses from 2013
- 9 Query traverses from 2012
- Velodyne HDL 32E LiDAR



Oxford Radar [2]

- 9km Route
- 2 Reference Traverses
- 5 Query traverses
- Velodyne HDL-32E LiDAR



WildPlaces [3]

- 33km across two distinct routes
- 12 traverse pairs per route
- Velodyne VPL 16 LiDAR



Natural Environment



Urban Environment

Place Recognition Performance

- **Urban environments:** Strong improvement over baseline on two datasets

Method	Oxford Radar	NCLT
M2DP	50.9	36.2
PN-VLAD	86.6	62.8
PCAN	80.3	59.0
LPD-Net	90.0	72.5
DH3D	78.2	59.4
BVMatch	93.9	83.6
Ours	94.6	92.9

Average Recall@1 on NCLT and Oxford Radar Dataset

- **Natural environments:** Outperforms baselines by ~15% showing robustness to terrain variation

Method	Venman	Karawatha	Average
Scan Context	33.98	38.44	36.21
TransLoc3D	50.24	46.08	48.16
MinkLoc3Dv2	75.77	67.82	71.80
LoGG3D-Net	79.84	74.67	77.26
Ours	94.46	90.50	92.48

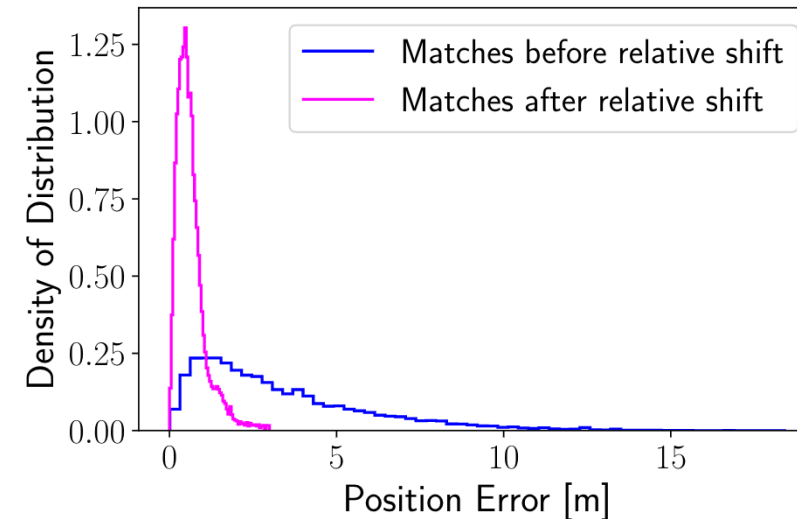
Average Recall@1 on WildPlaces Dataset

Relative Pose Estimation Performance

Pose Estimation Results on NCLT

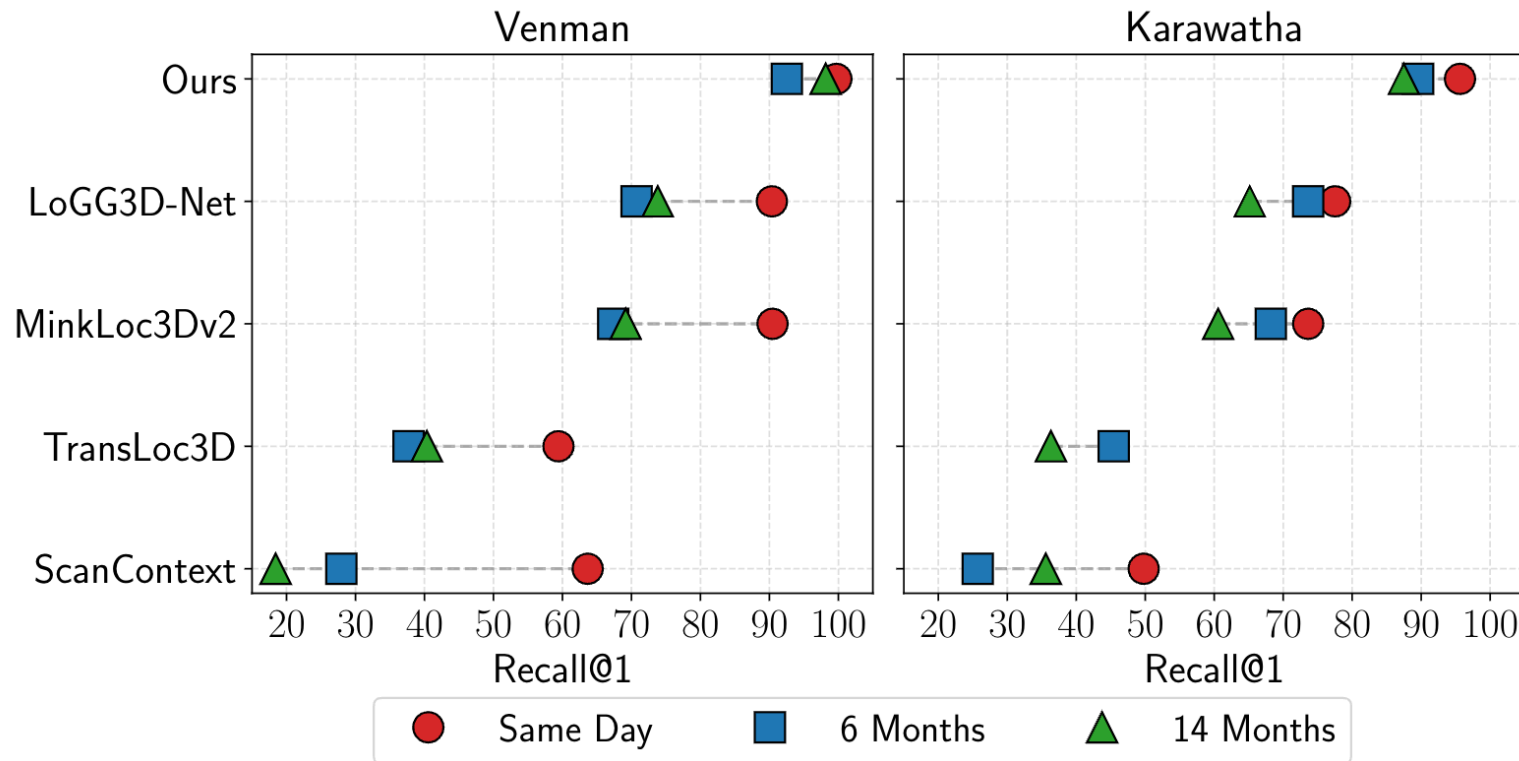
Method	RTE (m)		RRE (deg)		SR (%)
	Mean	Std.	Mean	Std.	
SIFT	0.69	0.46	1.49	1.23	53.3
DH3D	1.16	0.54	3.43	1.06	14.8
OverlapNet	-	-	2.43	1.42	15.4
BVMatch	0.57	0.38	1.08	1.00	94.5
Ours	0.48	0.29	1.43	1.07	97.7

- **Direct pose estimation** has lowest translation error and competitive rotational error —no ICP or RANSAC



- Correcting relative shift for the matched scan **reduced the variance in position error**, improving overall localisation

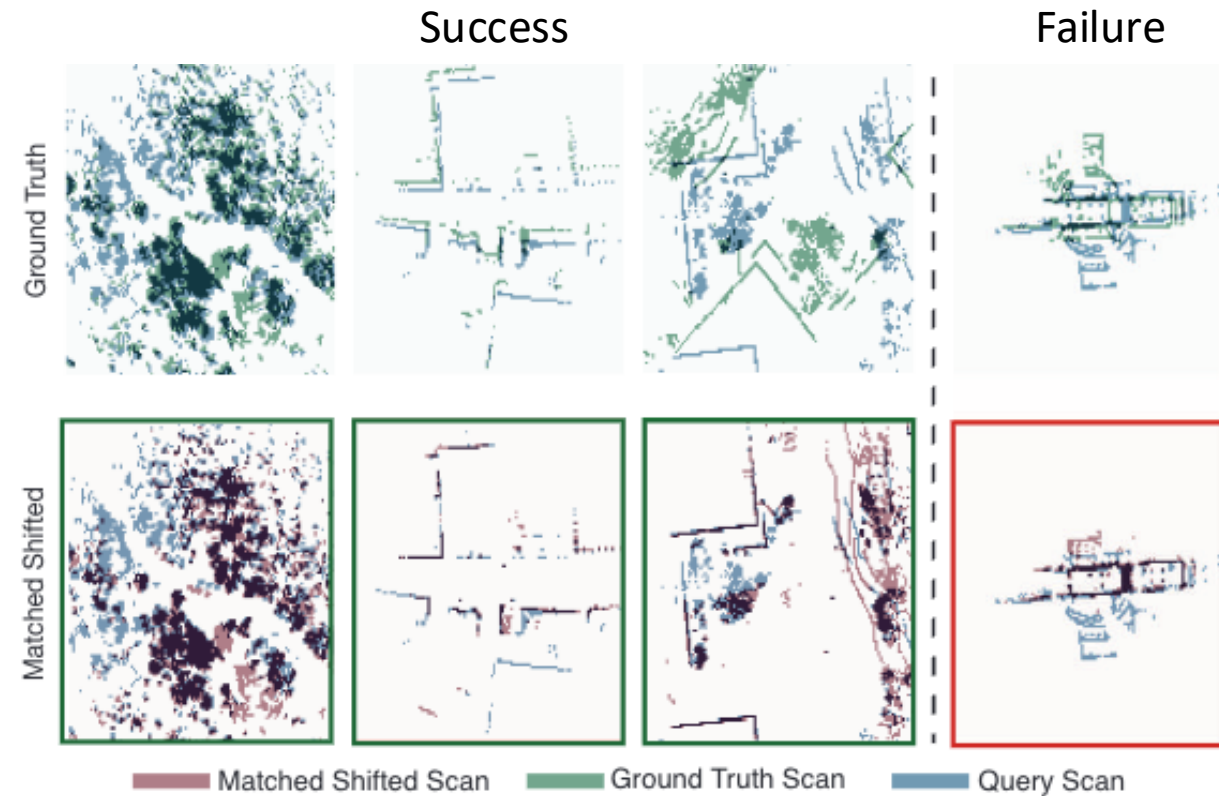
Long-term Performance



- Maintains **high recall over time**, even with 6- and 14-month environmental changes in WildPlaces

Limitations

- **Featureless environments** – Struggles in areas with repetitive structures, minimal distinctive features like narrow paths with uniform geometry
- **Coarse rotation increments** – May lead to small pose estimation errors; finer increments increase runtime
- **Scalability** – Search time increases with large reference maps; optimisation needed for longer traverses



Thankyou



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https://github.com/theresejoseph/Matched_Filter_based_LPR